MILITARY SPECIFICATION

TAPE, FAN-FOLDED

1. SCOPE

1.1 Scope.- This specification covers two types of fan-folded tape intended to be perforated for use with photo-electric readers.

1.2 Classification.- Tape covered by this specification shall be of the following types:

Type I	-	0.6875 inch wide, nominal
Type II	-	1.0 inch wide, nominal

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacturing and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Ordnance Materiel, Packaging of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

U.S. Army, Frankford Arsenal

F10531830	Reproducer, Signal Data AN/GSQ-64

(Copies of specifications, standards, and drawings required by the suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 1220

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2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply:

TECHNICAL ASSOCIATION OF THE PULP AND PAPER INDUSTRY (TAPPI)

Standard T 404m - 50	Tensile Breaking Strength of Paper and Paperboard
Standard T 414m - 49	Internal Tearing Resistance of Paper
Standard T 423m - 50	Folding Endurance of Paper
Standard T 446m - 48	Moldability (Crease Retention) of Wrapping Paper

(Application for copies should be addressed to the Technical Secretary, TAPPI), 360 Lexington Ave., New York City).

3. REQUIREMENTS

3.1 Order of precedence.- Should any conflict occur between the contract requirements, this specification or other material specifications, the order of precedence shall be as specified in MIL-F-13926.

3.2 Materials.

3.2.1 Toxicity.- Materials known to produce harmful toxic effects, which may be injurious to the unprotected user after repeated tape handling, shall not be used.

3.2.2 Lamination.- When laminated material is employed, there shall be no evidence of material separation.

3.3 Design and construction.

3.3.1 Flame resistance.- The tape shall not have a flash point of less than 850 degrees Fahrenheit (F).

3.3.2 Solubility.- The tape shall be insoluble in carbon tetrachloride, water, alchols, or hydrocarbons (gasoline and motor oils).

3.3.3 Grit content.- Tape shall not contain abrasive impurities in excess of .04 percent of the weight of the tape.

3.3.4 Resiliency.- A specimen of tape 3.0 by 0.5 inches shall have a resiliency of no less than 65%, where resiliency is equal to:

$$\frac{CL}{SL} \times 100$$

Where: CL = Cord length of an arc formed by a specimen no more than 10 minutes after being curved and held in a semicircle, having a one inch radius, for no less than 1 hour

SL = Length of specimen of tape material prior to bending

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3.3.5 Perforating quality.- The quality of the tape shall allow perforating to be cleanly sheared without lint or dust residues.

3.3.6 Opacity.- The tape shall have an opacity of at least 95% throughout its length (see 3.3.13), when irradiated by at least a 0.1 lumen of radiation at wavelengths between 0.5 micron and 1.1 microns.

3.3.7 Internal tear resistance.- Tape shall have a minimum internal tear resistance of 70 grams.

3.3.8 Stiffness.- The tape shall have the ability to resist any deformable deviation from the elastic displacement medium when subjected to repeated flexing (see 6.3.4).

3.3.9 Tensile strength.- The tape shall have a minimum tensile strength of 15 pounds in the longitudinal direction (see 6.3.2), with stress imposed at right angles to the fold.

3.3.10 Crease retention.- Crease retention of the tape shall be not less than 75% in the longitudinal direction, after a pressure of one pound applied at the fold of a previously unfolded specimen, for a minimum of 30 seconds. The average percentage crease (C) is equal to:

$$\frac{100\ (180 - A)}{180}$$

Where: A = average angle formed in the crease

3.3.11 Tape widths.- Type I tape shall be 0.6875 ± 0.007 inch in width throughout its entire length. Type II tape shall be 1.0 ± 0.007 inch in width throughout its entire length.

3.3.12 Tape edge.- The tape shall have clean cut longitudinal unbroken edges throughout its entire length.

3.3.13 Tape length.- Tape length shall be as specified in the contract and shall be without splice for its entire length.

3.3.14 Tape folds.- Tape shall be fan-folded throughout its entire length. The bend axis of each fold shall be 8.5 ± 0.03 inches from its adjacent axis and perpendicular to both longitudinal edges.

3.3.15 Tape thickness.- Tape shall exhibit a thickness of 0.004 ± 0.0004 inch throughout its entire length under a steady pressure of 0.40 to 0.63 kilograms per square centimeter.

3.3.16 Color.- Unless otherwise specified in the contract, the color of the tape shall be optional

3.3.17 Finish.- Tape shall have a hard smooth finish and a maximum freedom fron unbeaten fibers, fluff, lint or dust, slime-spots, tears, pinholes. Folds, other than those specified herein, shall not be exhibited (see 3.3.14).

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3.3.18 Weight.- The weight of the Type I finished tape shall be between 0.9 pound and 1.1 pounds per 1000 feet. The weight of the Type II finished tape shall be between 1.3 pounds and 1.4 pounds per 1000 feet.

3.4 Moisture resistance.- The tape shall show no evidence of damage nor exceed the tape width, length or thickness by 2%, due to absorption or drying out of moisture after exposure to the RH conditions specified (see 3.6.2). Absorbency shall not have potential of causing hardness when subjected to low temperatures nor deter its performance.

3.5 Fungus resistance.- The tape shall be inherently fungistatic or treated with a suitable fungicidal agent.

3.6 Climatic service conditions (see 6.3.4).- The tape shall be designed and constructed to withstand any probable combination of temperatures and relative humidity (RH) conditions specified below.

3.6.1 Temperatures.

a.	Operating:	-40°F to +135°F
b.	Nonoperating and storage:	-80°F to +160°F

3.6.2 Relative humidity.

a.	Operating:	Up to 97%, including condensation due to temperature changes.
b.	Nonoperating and storage:	Up to 97%, including condensation due to temperature changes.

3.7 Performance.

3.7.1 Folding and stacking.- Tape shall fan-fold, (see 6.3.3) and stack a minimum of 90 folds to the inch when fed into a trough at least one inch deep, at the rate of at least 50 inches per second. The top fold of the specified stack of 90 folds shall be no more than 5.25 inches below the exit point of the feed mechanism. The feed mechanism and trough shall be compatible to that used on the Signal Data Reproducer AN/GSQ-64 (Drawing F10531830).

3.7.2 Folding endurance.- Tape shall not sever when subjected to 1500 double folds in the machine direction (see 6.3.1) at a uniform rate of 175 ± 25 double folds per minute, with an applied tension of 2.2 pounds. Folding shall be an oscillation to both right and left of the position of zero fold. Movement shall be through an arc of 135° ± 5° lying on a plane perpendicular to the bending axis.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the

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performance of all inspection requirements as specified herein. Except as otherwise specified the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspection are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Inspection provisions.

4.2.1 Submission of product.- Unless otherwise specified herein, or by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of product" as specified in MIL-STD-105.

4.2.2 Sampling.- For the purpose of sampling, each individual tape length as specified in the contract shall be considered a unit of product.

4.2.3 Examination and tests.- Examinations and tests related to Section 3 herein shall be performed on a defect (individual characteristic) basis utilizing the inspection levels and sampling plans specified in MIL-STD-105 and the Classification of Defects in Table I. Examinations and tests for packaging, packing and marking shall be in accordance with MIL-P-14232 and Section 5 herein.

4.2.4 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

TABLE I - CLASSIFICATION OF DEFECTS

CRITICAL: NONE	REQUIREMENTS	TEST PROCEDURES
MAJOR: AQL 1.0 percent defective		
101. Tape width	3.3.11	4.5.2.10
102. Tape edge	3.3.12	4.5.2.11
103. Tape length	3.3.13	4.5.2.12
104. Tape folds	3.3.14	4.5.2.13
105. Tape thickness	3.3.15	4.5.2.14
106. Color	3.3.16	4.5.2.15
107. Finish	3.3.17	4.5.2.16
108. Weight	3.3.18	4.5.2.17

4.3 Standard sampling.

4.3.1 General.- One length of fan-folded tape shall be selected at random by a Government representative as a sample from each 100 lengths produced (see 3.3.13). The sample shall meet the requirements and tests in Table II.

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TABLE II

CHARACTERISTICS	REQUIREMENTS	TEST PROCEDURES
301. Toxicity	3.2.1	Certification
302. Lamination	3.2.2	4.5.3(b), 4.5.5 & 4.5.6
303. Flame resistance	3.3.1	4.5.2.1
304. Solubility	3.3.2	4.5.2.2
305. Grit content	3.3.3	4.5.2.3
306. Resiliency	3.3.4	4.5.2.4
307. Perforating quality	3.3.5	Certification
308. Opacity	3.3.6	4.5.2.5
309. Internal tear resistance	3.3.7	4.5.2.6

4.3.2 Environmental.- Three lengths (see 3.3.13) shall be selected at random by a Government representative as a sampling from each 500 lengths produced or from each month's production, whichever occurs first. The samples shall have met the requirements and tests in Table I and shall then meet the requirements and tests in Table III.

TABLE III

CHARACTERISTICS	REQUIREMENTS	TEST PROCEDURES
310. Operating temperatures	3.6.1(a)	4.5.5 & 4.5.7
311. Storage temperatures	3.6.1(b)	4.5.5
312. Relative humidity - Operating & Non-operating	3.6.2(a) & (b)	4.5.6, 4.5.2.4 4.5.2.7, 4.5.2.10, 4.5.2.12, 4.5.2.14 & 4.5.2.17

4.3.2.1 Fungus resistance.- One tape shall be randomly selected from the first months production for testing in accordance with 4.5.4. Following successful completion, subsequent acceptance for fungus resistance requirements shall be by certification.

4.3.3 Physical.- Three lengths (see 3.3.13) shall be selected at random by a Government representative as a sampling from each 500 lengths produced or from each month's production, whichever occurs first. The samples shall have met the requirements and tests in Tables I and III and shall then meet the requirements and tests in Table IV.

TABLE IV

CHARACTERISTICS	REQUIREMENTS	TEST PROCEDURES
313. Stiffness	3.3.8	4.5.2.7
314. Tensile strength	3.3.9	4.5.2.8
315. Crease retention	3.3.10	4.5.2.9
316. Moisture resistance	3.4	4.5.3
317. Folding and stacking	3.7.1	4.5.7
318. Folding endurance	3.7.2	4.5.8

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4.3.4 Failure of sample.- Should any one item of a standard sampling fail to meet the specified test requirements, acceptance of the represented production lot will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved (see 4.2.4).

4.4 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply equipment in accordance with the applicable requirements of MIL-I-45607. The Government reserves the right to use the test equipment for its own independent inspections to the extent that such use will not unduly interfere with the contractor's delivery schedule.

4.4.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, care and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.4.2 Contractor design.- The contractor shall provide inspection equipment compatible with the "Test Methods and Procedures" specified in 4.5 and with "Test facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5 Test methods and procedures.

4.5.1 Material.- The materials used in manufacture of the tape shall be examined to insure compliance with 3.2. In addition to the tests specified herein, data shall be available showing evidence of material analysis prior to acceptance by certification of those characteristics for which certification is required.

4.5.2 Design and construction.

4.5.2.1 Flame resistance.- Conduct this test in a draft free chamber having a temperature of 135° ± 4°F and utilizing an electrical energized soldering iron with a tip temperature of 850°F. Position the tip of the iron at the lower edge of the tape specimen and hold the tip against the tape until an area 1/4 inch square, is disintegrated by the heat and then withdraw the tip of the iron. Continue to observe the burned area for evidence of flame until "after glow" is completely extinguished to determine conformance with the requirements of 3.3.1.

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4.5.2.2 Solubility.- Perform this test by immersing 3" strips of the tape, accurately weighed to the nearest 0.01 gram, into each of the solutions specified in 3.3.2 for a period of 2 hours. At the end of this time period remove the solutions and wash the specimens in a water bath. Dry the specimens to a constant weight at a temperature of 105° ± 3°C. Accurately weigh the treated specimens and compare this weight with the tape dry weight before immersion to determine compliance with 3.3.2.

4.5.2.3 Grit content.- To determine the grit content of paper tape for compliance with 3.3.3, de-oil approximately 20 grams of composite sample in a suitable extraction apparatus, such as a Soxhlet or Underwriters, and extract with carbon tetrachloride until all oil is removed. Dry the de-oiled paper to constant weight at a temperature of 105° ± 3°C. Weigh out 10 grams of the air-dried extracted tape to the nearest 0.01 gram and place the weighed sample in a clean 800 - milliliter Kjeldahl flask. Add 100 milliliters of fuming HNO_3; warm until the paper becomes a thin paste and then boil gently until a low viscosity solution results. Evaporate the solution to a volume of 30-40 milliliters, cool, add 20 milliliters of concentrated H_2SO_4 and heat until white fumes of SO_3 are evolved. Clarify the solution by cautiously adding small volumes of fuming HNO_3, allowing the drops to run down the inside wall of the flask and reheat until SO_3 fumes are evolved. Repeat this clearing operation as often as necessary to produce a clear light straw color solution which does not darken after further heating. When clear, give solution a final fuming to remove the HNO_3. Cool the solution to room temperature and dilute to 400 milliliters with distilled water. Boil gently for several minutes, and filter through a weighed fine fritted glass crucible. Rinse the flask three times with distilled water and after each rinse pour the water through the fritted crucible. Dry the crucible with contents, and weigh. Determine compliance with 3.3.3 by calculating the percentage of grit to the nearest 0.001 percent based on the constant weight of paper dried at 105° ± 3°C.

4.5.2.4 Resiliency.- Prior to conducting this test the specimens shall be cut with clean and parallel edges to a length of 3 inches. Remove any pronounced curvature of the tape in the direction of their length by rubbing the 3 inch strip over a smooth edge before testing. Accurately measure the length of the specimen with standard measuring equipment and record the measurement. Wrap the specimen around a 2 inch mandrel and clamp the tape flat against the mandrel for one hour. At the end of this time period, remove the tape specimen from the mandrel. Within ten minutes after removal from the mandrel place the tape on a flat plate and accurately measure the chord length. Record this measurement. To determine compliance with 3.3.4 calculate the amount of resiliency in the tape specimen by substituting the recorded measurements for the applicable symbols in the formula specified in 3.3.4.

4.5.2.5 Opacity.- The tape shall be tested for opaqueness at 10 equally spaced points over the entire length of the tape by means of a spectrophotometer calibrated to produce a minimum of 0.1 lumen of radiation between the wave lengths of 0.5 microns and 1.5 microns or by means of a photo electric cell combined with a suitable filter system, tungsten light source, and a galvanometer connected to the photoelectric cell.

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The intensity of the tungsten light source shall be adjusted to produce the specified amount of radiation and wavelength either by rheostat to vary the voltage or by adjusting the position of the tungsten source. The photoelectric cell unit shall be operated in a darkroom or a suitable box finished on the inside in dull black to reduce reflection to a minimum. The box shall be fabricated to prevent any external light from reaching the photoelectric cell when making the measurements. With the photoelectric cell positioned in front of the exit aperture of the tungsten light source at the proper position to give the specified radiation, the initial reading shall be considered 100%. A reading of the galvanometer, taken with the tape positioned flat in front of the photoelectric cell, will give the percentage of light transmission through the tape. This percent figure shall be determined as:

$$R1 \times 100 = \% \text{ of light transmission.}$$

When:

R_1 = Reading on the galvanometer with the tape positioned in front of photoelectric cell.
R_2 = Initial reading on the galvanometer with the tungsten light source calibrated at the specified radiation.

Light transmission shall not exceed 5% to determine compliance with the requirements of 3.3.6.

4.5.2.6 Internal tear resistance.- Tear resistance of the tape material specimen shall be tested in accordance with Test Procedure T414m-49 of T.A.P.P.I. (see para 2.2). The force required to tear a single sheet shall not be less than the resistance value, in grams, as specified in 3.3.7.

4.5.2.7 Stiffness.- Prior to conducting this test the tape material specimens shall be cut with clean and parallel edges to 1-1/2 inches wide by 2 inches long. Eliminate any very pronounced curvature of the strips in the direction of their length or width by rubbing the strips over a smooth edge before testing. To determine compliance with 3.3.8, the tape material specimen shall exhibit a minimum stiffness value of 160 milligrams when tested with a Gurley stiffness tester.

4.5.2.8 Tensile strength.- The tensile strength of the tape shall be tested in accordance with Test Procedure T404m-50 of T.A.P.P.I. (see para. 2.2) except that the stress shall be applied at a uniform rate of 12 inches per minute. The tensile strength shall not be less than the poundage value specified in 3.3.9.

4.5.2.9 Crease retention.- Crease retention shall be tested in accordance with Test Procedure T446m-48, Method B, of T.A.P.P.I. (see para. 2.2) to determine compliance with 3.3.10.

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4.5.2.10 Tape width.- Standard measuring equipment shall be used to measure the applicable tape width to determine compliance with the requirements of 3.3.11.

4.5.2.11 Tape edge.- The tape shall be subjected to a visual and tactile inspection to determine compliance with the requirements of 3.3.12.

4.5.2.12 Tape length.- The completed fan folded tape shall be subjected to a visual and dimensional inspection to determine compliance with the requirements specified in 3.3.13. Dimensional inspection shall be performed using standard measuring equipment.

4.5.2.13 Tape folds.- The tape shall be subjected to a visual and dimensional inspection to determine compliance with the requirements of 3.3.14. Dimensional inspection shall be performed using standard measuring equipment.

4.5.2.14 Tape thickness.- Tape thickness shall be measured at 10 equally spaced points over the entire length of fan folded tape. Measurements of the tape shall be made by means of standard measuring equipment with pressure applied as specified in 3.3.15. The tape thickness at all points tested shall be within the tolerance specified in 3.3.15.

4.5.2.15 Color.- The tape shall be subjected to a visual inspection to determine compliance with the contract requirements as specified in 3.3.16.

4.5.2.16 Finish.- Visual and tactile inspection techniques shall be used to inspect the finished tape for compliance with the requirements of 3.3.17. The visual inspection shall be performed with the aid of at least a 3 power optical element.

4.5.2.17 Weight.- The weight of the finished tape and the basis weight of the material used to manufacture the finished tape shall be measured with standard measuring equipment. The weight shall be within the limits specified in 3.3.18.

4.5.3 Moisture resistance.- To determine the moisture resistance of paper tape for compliance with 3.4 dry the complete tape to a constant weight at a temperature of 105°C (221°F) ± 3°C. Remove two specimens from the complete tape. Specimen #1 shall consist of a minimum stack of 100 folds. Specimen #2 shall be 24 inches long. Each specimen shall be tested as specified below:

(a) Specimen #1.- Accurately measure the 24 inch long specimen as specified in 4.5.2.10, 4.5.2.12, 4.5.2.14 and 4.5.2.17 and record these measurements. The specimen shall then be subjected to the condition specified in 4.5.6(a) for a period of 4 hours. At the end of this time period accurately measure the specimen again as specified above

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and record the measurements. Dry the specimen to a constant weight at a temperature of 105°C (221°F) ± 3°C. Again measure the specimen as specified above and record the measurements. Compare the recorded measurements with the original measurements. The variance of all recorded measurements shall be within 2 percent of the measurements at the original dry weight for compliance with 3.4.

(b) Specimen #2.- Subject this specimen to the humidity condition specified in 4.5.6(a) for a period of 4 hours. Upon completion of exposure, remove the surface moisture from the specimen with a lint free dry cloth or paper towel. Within ten minutes after completion of this exposure subject the specimen to the low temperature procedure outlined in 4.5.5. Subsequent to the 4 hour exposure the tape shall be subjected to the test specified in 4.5.7. The tape shall be visually examined for evidence of physical damage such as cracking, tearing and lamination separation, during and after the specified test, for compliance with 3.4.

4.5.4 Fungus resistance.- This test shall be conducted as defined in MIL-F-13926 to determine compliance with the requirements specified in 3.5.

4.5.5 Temperature (storage and operational).- This test shall be conducted at the temperatures specified in 3.6.1 with a maximum humidity of 60%. The tape shall be visually examined for evidence of physical damage such as cracking, tearing and lamination separation during and after the specified exposures. While thermally stabilized at the temperature extremes specified in 3.6.1(a) the tape shall be subjected to the test specified in 4.5.7 to determine compliance with the requirements referenced therein. Upon completion of the tests at the specified exposures and return to standard ambient temperature (60° to 90°F) the tape shall be subjected to the tests specified in 4.5.2.4, 4.5.2.7, 4.5.7, and 4.5.8 and shall conform to the requirements referenced therein.

4.5.6 Relative humidity.- The tape shall be subjected to the following high and low humidity conditions using distilled or demineralized water having a ph value between 6.5 and 7.5 at 77°F to obtain the desired humidity specified in 3.6.2. The tape shall be placed in the test chamber so that it is no closer than 6 inches from the sides of the chamber and is not being subjected to radiant heat. Visually examine tape for evidence of physical damage such as cracking, tearing and lamination separation during and after the specified humidity conditons. Upon completion of the tests at the humidity conditions specified in (a) and (b) below, the tape shall be tested in accordance with 4.5.7 at standard ambient temperature (60° to 90°F).

(a) High humidity.- Place tape in test chamber and raise temperature to between 80°F to 85°F. The humidity of the chamber shall then be adjusted to between 92% and 97% with condensation and maintained at that value for a minimum of 4 hours. Within 10 minutes after completion

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of this exposure, subject the tape or specimen to the tests specified in 4.5.2.4, and 4.5.2.7.

(b) Low humidity.- Repeat the above procedure except that humidity shall be adjusted to a maximum of 15% for the 4 hour exposure.

4.5.7 Folding and stacking.- This test shall be performed by utilizing a feed mechanism and trough compatible to that used on the Signal Data Reproducer AN/GSQ-64. Machine feed a minimum stack of 100 folds of fan folded tape into the trough at the rate specified in 3.7.1 for three complete cycles. Standard measuring equipment shall be used to measure the thickness of 90 folds of tape and the length from the exit point of the feed mechanism to the top of the stack of 90 folds. Repeat measurements at the completion of each cycle to determine compliance with the requirements of 3.7.1.

4.5.8 Folding endurance.- To determine compliance with 3.7.2 the tape shall be tested in accordance with Test Procedure T423 M-50, Method M.I.T. of T.A.P.P.I. (see par. 2.2) using an applied tension of one kilogram. The tape shall be subjected to 1500 double folds at the rate specified in 3.7.2 without severing. Subsequent to the folding endurance test, the tape shall be subjected to a visual and tactile examination and shall show no evidence of physical failure.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.- Unless otherwise specified (see 6.2) tapes shall be packaged and packed in accordance with Level C requirements of MIL-P-14232.

6. NOTES

6.1 Intended use.- Tape covered by this specification, when perforated, will record a predetermined program. When passed through a photoelectric read head at a speed of 50 inches-per-second it will then fan-fold and stack in the required cannister, immune to adverse effects of electromagnetic or electrostatic charges. The punched perforations when converted into electrical pulses by the read-head will be used to perform desired functions, such as loading a magnetic memory with a predetermined program.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Applicable stock number
(c) Type of tape required (3.3.11)
(d) Length of individual tape (3.3.13); quantity of tapes
(e) Destination for shipment of tape(s)

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6.3 Definitions.

6.3.1 Machine or longitudinal direction (grain).- Direction of tape parallel to its intended forward movement. This is also termed "grain" and is at right angles to that termed "cross direction".

6.3.2 Fan-fold.- For the purpose of this specification, "fan-fold" is defined as accordion-pleated with the folds occurring every 8.50 ± 0.03 inches, 180° in alternately opposite directions, and perpendicular to the longitudinal edges of the tape.

6.3.3 Climatic service conditions.- The environmental climatic conditions for which the tape is designed and constructed to perform in compliance with its intended use set down in the design.

6.3.4 Deformable elastic medium.- In the case of a deformable elastic medium, stiffness is the ratio of a steady force to the elastic displacement produced by it.

Custodian:

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Preparing activity:

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